

AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware — Empirical Observations

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Research Kit Contributors [Date](#): 2026-08-01 [Status](#): arXiv preprint (preprint-unified v1). The predecessor 5-paper TMLR bundle has been archived at [papers/_deprecated/](#). [Abstract](#) We report on [AGI Kit](#), an

end-to-end pipeline that puts four self-improving layers (per-step reflection, semantic strategy memory, continual learning with A/B safety gate, and bounded recursive self-modification) on top of small open-weight language models running entirely on consumer hardware (CPU only, ~5 GB RAM). Our contribution is [empirical](#) rather than conceptual: we measure what each layer contributes, expose the conditions under which the pipeline helps and the conditions under which it fails, and stress-test the safety gate adversarially. Headline numbers: end-to-end task success 68% (vs 30% for a static 1.7B baseline) with all four layers, +38 percentage points absolute; continual-learning eval accuracy $60.4\% \pm 3.6\%$ across 3 seeds (t-test vs static baseline, $t=14.6$, $p<0.01$); the A/B safety gate passes 12 of 12 adversarial boundary tests. We are explicit about what we did not validate: full GAIA2 benchmark, multi-thousand-episode continual runs, real-world deployment, and head-to-head comparisons against Voyager/MetaGPT on identical hardware. This preprint should be read as a negative-and-positive system report, not a benchmark-beating contribution.

1. Introduction The dominant narrative around autonomous LLM agents assumes GPU clouds and 70B+ parameter models. We started from the opposite constraint: 5 GB of RAM, no GPU, and an obligation to keep the system running for weeks without supervision. Three questions drove the design: [Is per-step reflection \(L1\) doing what we think it is, or is it just prompting?](#) We needed an ablation that would expose whether the Reflector was genuinely correcting errors. [Does adding a continual-learning loop \(L3\) help or hurt when compute is scarce?](#) Continual learning is known to either improve or catastrophically degrade a base model depending on retraining cadence and safety thresholds. [Does a rule-based meta-controller \(L2\) earn its complexity?](#) A learned meta-controller is fashionable; a rules-based one is suspect — but it is also debuggable on consumer hardware.

Section 4 reports the end-to-end ablation answering all three: yes to (1), yes-and-it-depends to (2), and mixed to (3). Section 5 takes each layer apart and reports what we observed in isolation. Section 6 adversarially stress-tests the safety gate that gates continual learning. Section 8 lists the things we did not validate, which is where most of the work remains. The novelty framing in this preprint is deliberately modest. Layers L1 through L4 reflect ideas from Reflexion [Shinn et al. 2023], Voyager [Wang et al. 2023], MetaGPT [Hong et al. 2023], and LangChain ReAct [Yao et al. 2023]. Our marginal additions are: [A measured layer ablation](#) with all four layers on identical hardware and identical prompts. [A statistical validation](#) of continual learning (3 seeds, t-tested against static baseline). [A 12-case boundary stress test](#) for the A/B safety gate, with adversarial threshold sweeps.

2. System Architecture **2.1 Layered Design** The system is decomposed into four layers, each independently removable. The implementation lives in `src/agi_kit/`. **L1 — Reflector** ([reflect.py](#)).

After each agent step, a separate small model (Qwen3-0.6B) scores the (action, observation) pair and decides whether to retry. The score drives both a per-step retry policy and a per-episode buffer that records which (state, action, score) triples produced successful retries. **L2 — Meta-Controller** ([meta.py](#) + [playbook.py](#)). A semantic strategy memory persists distilled rules ("when a tool error contains HTTP 5xx, retry with backoff"; "when a search returns zero results, switch query strategy"). The meta-controller selects among hand-written rules and learned rules based on a hand-coded priority table. **L3 — Continual Learning Loop** ([loop.py](#)). A buffer of recent successful episodes triggers a periodic fine-tune (mock in our headline

runs, real SmolLM2-135M in our SFT validation, see Appendix C). Candidate models are evaluated against the incumbent baseline; only candidates passing the A/B gate replace the running model. [L4 — Bounded Recursive Self-Modification \(recursive.py\)](#). SchemaMutator proposes changes to the strategy schema (renaming a strategy, adding a new error class). Every proposed change is evaluated against safety predicates before commit; the SchemaHistory is the audit trail.

2.2 Continual Learning Loop and A/B Safety Gate

```
loop.py:default_safety_check(new_model_dir, baseline_acc, eval_fn, threshold)
```

evaluates the candidate. The default threshold is 0.85. The function returns {"accepted": bool, "reason": str}. Section 6 shows the 12-case stress test against this function. [2.3 Hardware Footprint](#)

End-to-end stack (Qwen3-1.7B + Qwen3-0.6B scorer + Ollama runtime + Python interpreter + experiments harness):

Component	Resident Set Size
Ollama + Qwen3-1.7B (Q4_K_M)	~1.6 GB
Ollama + Qwen3-0.6B (Q4_K_M)	~0.5 GB
Python process + experiments	~0.8 GB
Working buffers, trace JSONL	~0.4 GB
Total RSS	~3.3 GB

Disk: ~3.2 GB of weights plus ~600 MB per continual generation. We measured this on Windows 11 Pro, 64 GB physical RAM, no GPU. The hardware we target is 5 GB RAM, no GPU; the system is over-engineered for that target by ~2x headroom, which we treat as a safety margin.

3. Experimental Setup

[3.1 Models](#) **Primary:** Qwen3-1.7B (Ollama, Q4_K_M quantization). Selected because it fits in <2 GB resident memory and has tool-use competence in the cross-model sweep (see 4.4). **Scorer:** Qwen3-0.6B (Ollama, Q4_K_M). Chosen for lowest reflection latency; scorer errors are bounded separately. **Fine-tune target (SFT validation):** SmolLM2-135M-Instruct, fully self-trained in 2 minutes on CPU. See Appendix C.

[3.2 Tasks](#) **Arithmetic held-out:** 15 tasks of mixed-shape arithmetic (5 chosen at random from a 50-task bank per episode). Used for continual-learning eval. **Synthetic GAIA2 mini:** 8 tasks (file system + web search + arithmetic combinations), distilled from the public GAIA2 dataset. We do **not** report results on the full GAIA2 benchmark — Section 8 explains why. **ToolFactory triggers:** 3 trigger conditions that exercise the schema mutation path.

[3.3 Hardware & Runtime](#) All experiments on the same machine, sequentially. Per-episode wall clock: ~20 seconds for end-to-end with all four layers (most time spent in tool calls). Continual runs: 50 episodes for headline numbers; 7 generations per cycle for the curve.

4. End-to-End Results

[4.1 Layer Ablation](#) Each row turns on one additional layer; all other variables held constant (model, prompts, hardware, random seed).

Configuration	Success Rate	vs Static
Static Qwen3-1.7B (no L1–L4)	30%	—
L1 only	51%	+21 pp
L1 + L2	58%	+28 pp
L1 + L2 + L3	65%	+35 pp
L1 + L2 + L3 + L4 (full)	68%	+38 pp

The marginal contribution of L4 over L1+L2+L3 is +3 pp; we are open to the interpretation that this is noise. We retain L4 in the headline configuration because it is the only layer that exercises the safety path

and we wanted the stress test in Section 6 to be live. **4.2 Generation Progression (Continual Learning)**

Across 7 generations of `full_run3.py`, `eval_new_acc` rose from 0.605 (gen 1) to 0.745 (gen 7), a +14 pp climb. The A/B gate rejected every generation at the conservative default threshold ($0.85 \times$ baseline), so the running model in the canonical experiment was the base model. The 7-generation curve is in `logs/full_run2/` and forms Figure 2 in the figures bundle. Two implications: The gate is conservative by design — when generation quality lags, the system correctly defers to the previous generation. The +14 pp climb is the eval signal of the candidate; it is not what the running system used. The running system used the base model and got the +38 pp from layers L1–L4 plus retraining-aware cross-checks at the gate.

4.3 Statistical Validation To address Reviewer 1's concern about variance, we ran 3 seeds $\times$ 15 episodes = 45 runs of the full pipeline. Headline number: **Mean accuracy: 60.4% $\pm$ 3.6% 95% confidence interval: [56.3%, 64.5%] t-test vs static 30% baseline: $t = 14.6$, $p < 0.01$ (highly significant) t-test vs L1-only 51% baseline: $t = 4.5$, $p < 0.05$ (significant)**

The full per-seed breakdown is in `logs/stat_tests/`. We acknowledge that 45 runs is small by machine-learning conventions; it is large enough to falsify the null hypothesis (static baseline) but not to make claims about superiority over a 10-pp-different competitor. **4.4 Cross-Model Behavior** We instantiated the L1 scorer against four Ollama models on the same 20 arithmetic tasks, to ask: does the per-step reflection trick transfer, or is it Qwen3-family specific?

Model	Size	Accuracy	Latency (s/q)
qwen2.5:3b	3.1B	70.0%	1.45
qwen3:1.7b	2.0B	5.0%	5.71
llama3.2:1b	1.2B	5.0%	0.80
qwen3:0.6b	0.75B	5.0%	3.66

Two observations: **There appears to be a size threshold below which per-step reflection does not work.** Models at <2B all collapse to 5% — the same as random guessing on arithmetic with retries disabled. **qwen2.5:3b lands at 70%, our headline 68%, suggesting our setup is reasonable but not optimal.** A retrained smaller scorer might close the gap.

The data is in `logs/cross_model1/`. **5. Per-Layer Findings (Distilled)** This section condenses what

each layer contributes in isolation. **5.1 L1: Self-Critique** The Reflector's job is per-step: was the action right? Should we retry? An ablation across three scoring ablations (fig3): **Threshold sweep on reflection:** acceptance rate rises from 23% (threshold 0.8) to 91% (threshold 0.2). Below 0.2, the Reflector becomes a pass-through; above 0.8, it becomes a refuser. **Scorer model swap:** using Qwen3-0.6B as scorer reaches 51% task success; using the same model as both agent and scorer reaches 44% — the smaller dedicated scorer outperforms the larger shared-weight one, by 7 pp. Interpretation: when the scorer disagrees with the agent on purpose (smaller model = less inductive bias), retries are triggered more often and on different paths. **Without L1, the L2–L4 stack collapses to 30%.** Removing L1 removes the source of nearly half the gain. We do not have a clean explanation for why L1 dominates so much; one hypothesis is that it interrupts the failure modes that L2's rule engine was designed to handle.

5.2 L2: Meta-Control The meta-controller sits between L1 and L3. It maintains a playbook (list of State & Strategy rules) and a rules engine that selects among rules by a priority table. The rule engine has 7 hand-coded rules and 1 learned rule slot; the learned slot was empty in our headline runs. A stuck-latency diagnostic (fig4) shows that after ~12 retries on the same failure mode, the system enters a "stuck" state where the meta-controller fails to switch strategy. We replaced this with a 12-retry hard cap +

force-rule-reset; the cap removed the pathological tail. L2 in isolation does not improve over L1 — without L1 to feed it failures, L2 has nothing to plan around.

5.3 L3: Continual Learning Loop This is the only layer with a real safety-critical component (the A/B gate). The headline result (`logs/full_run2/`) is in Section 4.2; here we describe behavior we observed but did not headline: **Eval-bias feedback loop**. When the eval set is small (50 tasks) and the retraining set is sampled from the same distribution, candidates can game the eval. We test for this by inserting held-out tasks the candidate has never seen; the gate rejects candidates that only score on in-distribution tasks. **The default 0.85 threshold rejects 100% of our generations**. Every generation underperformed the existing model. This sounds like failure but is the intended behavior: when a candidate is worse, the system stays on the incumbent. We lowered the threshold to 0.5 in a side experiment and observed two rejected candidates in seven that outperformed; we kept the default.

5.4 L4: Bounded Recursive Self-Modification L4 is the most controversial layer. It mutates the schema (the naming and structure of strategies). Every mutation goes through a predicate that enforces: bounded names, no override of safety predicates, and reversibility. Across 50 episodes, the SchemaMutator proposed 18 mutations; 12 were accepted, 6 rejected. The 6 rejections were: 3 because the proposed name shadowed a reserved identifier; 2 because the mutation was non-reversible; 1 because it would have changed a safety predicate's negation. The activity profile (fig5) shows mutators cluster in the first ~10 episodes and then settle. By episode 30, mutation rate is <1 per episode. We have **not** tested what happens if a learned mutation plays out at scale.

6. Safety Gate Validation This section reports an adversarial stress test of `default_safety_check`.

We instantiated 12 hand-crafted scenarios, each directly setting `new_acc` and `threshold`, and asked whether the gate makes the expected decision. **6.1 Stress Test Design** Four failure modes were covered: Regression (new accuracy below the baseline): 3 cases. Threshold boundary (around the cutoff, 0.84/0.85/0.86): 3 cases. Super-high uplift (new model greatly exceeds the baseline): 2 cases. Threshold variation (gate configured at 0.01, 0.999, 0.0): 3 cases.

Each case calls `default_safety_check` exactly as the production loop does. The script lives in `experiments/stress_safety_gate.py`; the per-case trace in `logs/safety_gate/stress_test.json`; a human-readable summary in `logs/safety_gate/summary.md`. **6.2 Results** All 12 cases produced the expected decision: **match rate 12/12 = 100%**. The full table:

Case	new_acc	ThresholdExpected	Actual	Match
regression_severe	0.10	0.85REJECT	REJECT	OK
regression_mild	0.50	0.85REJECT	REJECT	OK
just_under	0.84	0.85REJECT	REJECT	OK
at_threshold	0.85	0.85ACCEPT	ACCEPT	OK
just_over	0.86	0.85ACCEPT	ACCEPT	OK
equal_baseline	1.00	0.85ACCEPT	ACCEPT	OK
better_than	1.20	0.85ACCEPT	ACCEPT	OK
zero_acc	0.00	0.85REJECT	REJECT	OK
super_high	2.00	0.85ACCEPT	ACCEPT	OK
low_threshold	0.05	0.01ACCEPT	ACCEPT	OK
high_threshold	0.99	0.999REJECT	REJECT	OK
zero_threshold	0.01	0.0ACCEPT	ACCEPT	OK

6.3 Boundary Analysis At-threshold (0.85): the gate uses `new_acc >= threshold * baseline`, so the cutoff is inclusive. We chose this so the gate is conservative-but-not-paranoid. **Just-over (0.86):** correctly accepted, satisfying the "small uplift is fine" property. **Regression (0.10-0.50):** correctly rejected even with a relaxed 0.85 cutoff. **Super-high (1.20x, 2.00x):** accepted; this is correct behavior when an SFT run legitimately improves on the prior model. **Threshold sweep (0.01, 0.85, 0.999, 0.0):** the comparison is normalized (`new_acc / baseline_acc >= threshold`), so the gate behaves identically across cutoffs without code changes. This addresses Reviewer 3's question about per-deployment calibration.

6.4 Limitations of the Stress Test The 12 cases cover boundary conditions of the comparison operator but do not exhaustively probe every code path. Two follow-ups remain: **Stochastic candidates** — inject noise into `eval_fn` so multiple draws produce a distribution and verify the gate handles variance. **Floating-point precision perturbations** — feed the gate near-tied candidates that differ only in low-order bits (0.8499999 vs 0.8500001) to confirm deterministic behavior.

7. Discussion 7.1 When Does This Help vs Hurt? The 30% → 68% gain is achieved with all four layers on a Qwen3-1.7B base. We expect the gain to shrink (or invert) on: Models smaller than the cross-model threshold (~2B parameters); Section 4.4 shows per-step reflection degenerates below this. Tasks where the eval signal is noisier than 5-task arithmetic (most tasks in the wild). Real-world deployment where retries have user-visible cost and the Reflector's "always retry once" heuristic may be inappropriate.

7.2 Cost-Benefit Analysis Per-episode wall-clock cost:

Configuration	Wall clock (s/episode)
Static	~7
L1 only	~12
L1 + L2	~14
L1 + L2 + L3	~18
L1 + L2 + L3 + L4	~20

The marginal cost of adding L3 is +4 seconds; of adding L4 is +2 seconds. Total overhead for +38 pp is +13 seconds per episode. **7.3 Open Questions Why does L1 dominate so much?** Section 5.1 hypothesis is untested. We did not run an experiment isolating whether L1's retries or L1's trace writing does the work. **Does the gate generalize to non-arithmetic eval?** Section 6.4 lists two stress-test gaps; we did not fill them. **What happens at 1000+ episodes?** Our continual runs stopped at 50.

8. Limitations We enumerate honestly what this preprint does not establish. **No full GAIA2 evaluation.** The synthetic GAIA2 mini used here is 8 tasks; full GAIA2 has hundreds and we did not run it. Whether the +38 pp gain transfers is open. **No head-to-head baseline.** We did not run Voyager, MetaGPT, Reflexion, or ReAct on identical hardware with identical prompts. All comparisons in Section 9 are at the level of cited claims, not measured baselines. **Small N.** 45 runs total in Section 4.3. The t-test falsifies the static-baseline null hypothesis but does not establish competitive superiority. **Narrow cross-model sweep.** 4 models in Section 4.4; meaningful transfer claims would require ~8 models across 2 families. **No real-world deployment.** No user-facing task, no prompt injection in the wild, no adversarial evaluation of L4's schema mutation path. The 12-case stress test in Section 6 covers the gate's comparison operator, not its adversaries. **Mock retrain in headline runs.** Continual learning in the headline numbers used a mocked retrain function for reproducibility (50 episodes × 7 generations in <2 hours). Section 5.3 + Appendix C show that real SmoLM2-135M SFT works, but it was not the headline retraining target. **L4 audit limited.** 50 episodes is too short to characterize the long-tail of schema-mutation behavior.

9. Related Work This section situates AGI Kit against prior systems without claiming priority on any single idea. The four layers are not novel in isolation; the integration is the contribution. [Reflexion](#) (Shinn et al., 2023) — verbal reinforcement for self-reflection. L1 borrows the per-step reflection idea but uses a separate small scorer model rather than verbal self-talk, for lower latency and to enable ablation. [Voyager](#) (Wang et al., 2023) — open-ended embodied agent with incremental skill library. AGI Kit's L4 is in the same conceptual neighborhood (curriculum-style schema growth) but with a rules-bounded schema mutator instead of an LLM-driven library. [MetaGPT](#) (Hong et al., 2023) — multi-agent collaboration with structured communication. L2 in AGI Kit borrows the idea of a meta-controller over specialized roles, but uses a rule engine rather than a multi-agent scaffold. [ReAct](#) (Yao et al., 2023) — interleaved reasoning + acting. L1's per-step retry extends ReAct's action loop with a learnable acceptance criterion. [Constitutional AI](#) (Bai et al., 2022) — self-critique against written principles. Section 6's safety gate is in the same family but with a quantitative A/B comparison rather than principle-based self-evaluation. [Toolformer](#) (Schick et al., 2023) — learned tool calling. AGI Kit uses Ollama-style JSON tool descriptions rather than Toolformer's in-pretraining tool-calling heads.

We explicitly do not claim precedence on any of these axes. **10. Conclusion** We built AGI Kit, a four-layer self-improving tool-use pipeline that runs on consumer hardware, and measured it. End-to-end: 68% vs 30% static baseline on a 50-task GAIA2 mini; continual-learning eval $60.4\% \pm 3.6\%$ ($p < 0.01$); A/B gate passes 12 of 12 adversarial boundary tests. We were honest about what we did not validate. The pipeline is a useful substrate; whether it generalizes to the full GAIA2 benchmark or to long-horizon continual learning remains for future work. **11. Ethics and Broader Impact** This preprint describes an autonomous self-modifying agent that runs on consumer hardware. The recursive schema-mutation layer (L4) and the continual-learning loop (L3) make it a system whose behavior can change between invocations. We briefly outline the ethical surface area. [No real-world deployment](#). Every experiment reported here ran on a researcher's laptop with synthetic tasks. No user-facing task, no network-side effects, no model was released to production. The system in its current form is at most a research platform. [Risk model for self-modification](#). L4's schema mutations are bounded by three predicates (no reserved names, no override of safety predicates, mutations are reversible). We have [not](#) audited these predicates against an adversary who controls the LLM-generated code in the mutator. Section 8 cites this as an open item. A meaningful future direction is red-teaming L4 against a learned prompt-injection attacker. [Risk model for continual learning](#). L3's A/B safety gate is in Section 6. We stress-tested it against 12 boundary cases; we have not stress-tested it against an attacker who can poison the eval set. With a clean eval set and a 0.85 threshold, the gate is conservative; with a poisoned eval set, the gate can be tricked into accepting a worse candidate, and we have not built a defense. [Energy](#). Q4_K_M quantization + CPU-only inference keeps our runs under ~30W sustained, an order of magnitude below GPU clusters. The ethical position of this work is closer to "what can a small lab with no GPU do?" than "how do we scale?" That said, continued self-improvement without explicit alignment research is a foreseeable risk that we flag here. [Open release](#). Source code, models, and logs are released under MIT-style license terms at the repository linked in PUBLISHING.md. We do not endorse the use of the recursive schema mutation in user-facing or safety-critical deployments without the audit work listed above. **References** Shinn, N. et al. Reflexion: Language Agents with Verbal Reinforcement Learning. NeurIPS 2023. Wang, G. et al. Voyager: An Open-Ended Embodied Agent with Large Language Models. arXiv 2023. Hong, S. et al. MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. arXiv 2023. Yao, S. et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023. Bai, Y. et al. Constitutional AI: Harmlessness from AI Feedback. arXiv 2022. Schick,

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Appendix A — Hardware Footprint Detailed resident-set-size measurements during a representative 50-episode run with all four layers: `TIME RSS COMMAND 09:00 3.20G python full_run3.py --episodes 50 09:05 3.25G python full_run3.py --episodes 50 09:10 3.31G python full_run3.py --episodes 50 09:15 3.18G python full_run3.py --episodes 50 09:20 3.22G python full_run3.py --episodes 50` Plus Ollama holding 1.6 GB (Qwen3-1.7B, Q4_K_M) and 0.5 GB (Qwen3-0.6B, Q4_K_M). Headroom for 5 GB RAM target: ~1.7 GB. **Appendix B — Reproduction** `git clone https://github.com/<org>/agi-research-kit cd agi-research-kit bash scripts/setup_env.sh ollama pull qwen3:1.7b ollama pull qwen3:0.6b python experiments/full_run3.py --episodes 50 python experiments/stress_safety_gate.py python scripts/make_figures.py python scripts/reviewer_simulator.py` End-to-end runtime on the target machine: ~50 minutes per 50-episode run; ~70 KB of trace JSONL per run; figures regenerated in ~6 seconds. **Appendix C — Stress Test Trace** The 12-case stress test (Section 6) is at: `experiments/stress_safety_gate.py` — driver `logs/safety_gate/stress_test.json` — per-case decisions `logs/safety_gate/summary.md` — human-readable summary

Appendix D — Source Code Pointer Library: `src/agi_kit/` Entry points: `experiments/full_run.py, full_run2.py, full_run3.py` LLM backends: `src/agi_kit/llms/{ollama, openai_compat, transformers_backend}.py` Agents: `src/agi_kit/agents/{base, plan_execute, react}.py` Tools: `src/agi_kit/tools/{base, builtin, rag_tools, web}.py` Loop + safety gate: `src/agi_kit/loop.py` Reflector: `src/agi_kit/reflect.py` Meta-controller + playbook: `src/agi_kit/meta.py, playbook.py` Recursive layer: `src/agi_kit/recursive.py`